

**Stanley College of Engineering & Technology for Women (A)****B.E (Common to all) II Semester (Main) Examinations-September-2023****Mathematics-II****Time: 03Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any four out of remaining six questions.  
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question Paper.  
 iii. Missing data, if any, may be suitably assumed.

**Part-A****6X2=12M**

1. a. Solve  $(y^2 - 2xy) dx + (2xy - x^2) dy = 0$  [2M CO1BTL2]  
 b. Solve  $(D^3 - 3D + 2) y = 0$  [2M CO2 BTL3]  
 c. Evaluate  $\int_0^1 x^5(1-x)^3 dx$  [2M CO3 BTL4]  
 d. Define Rank of Matrix with example. [2M CO4BTL1]  
 e. Find  $L \{ t^2 \sin 2t \}$  [2M CO5 BTL3]  
 f. Find the value of  $k$  such that the rank of  $\begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}$  is 2. [2M CO4 BTL2]

**Part-B****4X12=48M**

2. a) Solve  $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$  [6M CO1 BTL3]  
 b) Prove that the system of parabolas  $y^2 = 4a(x+a)$  is self-orthogonal. [6M CO1 BTL4]
3. a) Solve  $(D^2 + 3D + 2) y = 2 \cos(2x+3) + 2e^x + x^2$  [6M CO2 BTL4]  
 b) Solve the differential equation  $(D^2 + 4) y = \sec 2x$  by the method of variation of parameters [6M CO2 BTL3]
4. a) State and Prove relation between Beta and Gamma functions [8M CO3 BTL4]  
 b) Evaluate  $\int_0^\infty x^2 e^{-x^2} dx$  [4M CO3 BTL3]
5. a) Determine whether the following equations have a non-trivial solution, if so solve them:  $4x + 2y + z + 3w = 0$ ,  $6x + 3y + 4z + 7w = 0$ ,  $2x + y + w = 0$  [6M CO4 BTL3]  
 b) Verify Cayley-Hamilton theorem for  $A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$  [6M CO4 BTL4]
6. a) Evaluate  $L \left\{ \frac{\cos 2t - \cos 3t}{t} \right\}$  [4M CO5 BTL3]  
 b) Evaluate  $L^{-1} \left\{ \frac{s^2}{(s^2+4)(s^2+25)} \right\}$  using convolution theorem. [8M CO5 BTL4]
7. a) Solve  $P = \sin(y - xp)$ . Also find its Singular solution. [4M CO2 BTL2]  
 b) Find the nature of the quadratic form  $2x^2 + 2y^2 + 2z^2 + 2yz$  [8M CO4 BTL3]

Hall Ticket No. -

Code: R123931

**Stanley College of Engineering & Technology for Women (A)**

**B.E (ECE/EEE/IT) II Semester (Main) Examinations-September-2023**

**Essence of Indian Traditional Knowledge**

**Time: 03 Hours**

**Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.  
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.  
iii. Missing data, if any, may be suitably assumed.

**Part-A**

**6X2=12M**

1. a. "Culture is diverse" Explain. [2M CO1BTL1]  
b. Who is the author of "Pruthviraj Raso" and what is its specialty? [2M CO2 BTL1]  
c. Write about Ramakrishna mission. [2M CO3 BTL2]  
d. Explain how Kalighat painting got the name. [2M CO4 BTL2]  
e. Name any four women scholars of Vedic era. [2M CO5 BTL1]  
f. What is the meaning of the Sanskrit word "Itihasa". [2M CO2 BTL1]

**Part-B**

**4X12=48M**

2. a) What is unique about Indian Culture? [6M CO1 BTL2]  
b) What is the impact of culture on Human life? [6M CO1 BTL2]
3. a) Explain the five characteristics of Puranas. [6M CO2 BTL1]  
b) Write about the Telugu language and literature. [6M CO2 BTL2]
4. a) What were the religious beliefs of Pre-Vedic and Vedic people? [6M CO3 BTL2]  
b) Assess the contribution of Arya-Samaj. [6M CO3 BTL2]
5. a) Write a short note on Medieval Indian Painting. [6M CO4 BTL1]  
b) Write a short note in Indian classical music. [6M CO4 BTL1]
6. a) Describe the salient features of the NEP 2019. [6M CO5 BTL1]  
b) Discuss the development of Science and Technology in Modern India. [6M CO5 BTL2]
7. a) What is unique about the Indian Architecture? [6M CO4 BTL2]  
b) Write a short note on Ancient Indian Culture. [6M CO1 BTL1]

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Hall Ticket No. -

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**Stanley College of Engineering & Technology for Women (A)**

**B.E (ECE/EEE/IT) II Semester (Main) Examinations-September-2023**

**Indian Constitution**

**Time: 03 Hours**

**Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.  
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.  
iii. Missing data, if any, may be suitably assumed.

**Part-A**

**6X2=12M**

1. a. Write about Preamble. [2M CO1 BTL2]
- b. Explain about qualifications of President. [2M CO2 BTL3]
- c. Explain about Right to Equality of Indian Constitution. [2M CO3 BTL1]
- d. Write a brief note on Central-State relations. [2M CO4 BTL3]
- e. What is the Composition of NCW? [2M CO5 BTL2]
- f. Why do we need Voting Right in Democracy? [2M CO1 BTL3]

**Part-B**

**4X12=48M**

2. a) What are the Features of Govt of India Act 1909? [6M CO1 BTL1]
- b) What was the composition of Constituent Assembly? [6M CO1 BTL2]
3. a) Explain about President of India. [8M CO2 BTL3]
- b) Explain about Governor of India. [4M CO2 BTL2]
4. a) Explain about Fundamental Rights. [8M CO3 BTL2]
- b) Explain the difference between Fundamental Rights and Duties. [4M CO3 BTL3]
5. a) Explain about Relations between Union and State. [5M CO4 BTL2]
- b) Explain the Composition and Objectives of NITI Aayog. [7M CO4 BTL3]
6. a) Explain the Powers and Functions of Election Commission of India. [8M CO5 BTL3]
- b) Explain about Powers of National Commission for Women. [4M CO5 BTL1]
7. a) Write an essay on the Salient Features of Indian Constitution. [8M CO1 BTL3]
- b) Explain about 74<sup>th</sup> Constitutional Amendment Act. [4M CO2 BTL2]

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## Stanley College of Engineering &amp; Technology for Women (A)

B.E(ECE,EEE,IT) II Semester (Main) Examinations-September-2023

English

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.  
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.  
 iii. Missing data, if any, may be suitably assumed.

## Part -A

6X2=12M

2M L1 CO1

## 1. Do as directed.

- a. i) I am \_\_\_\_\_ university student. ( an/a/the/no article)  
 ii) The teacher congratulated Ravi \_\_\_\_\_ his success. (Write a suitable preposition.)  
 iii) \_\_\_\_\_ in the workplace is harmful to its growth. [Root word : Loyal] (Write the apt word by using suffix-prefix)  
 iv) Leela is the brightest student among all. (Change from superlative to positive degree of comparison)
- b. i) The boy is not meritorious. \_\_\_\_\_ he is very hard working. 2M L3 CO2  
 (Use suitable transition word)  
 ii) \_\_\_\_\_ he is very famous, Azim Premji's modesty is exemplary.  
 (Use an apt connective.)  
 iii) Identify the root words in the given blended expression **Infotainment**.  
 iv) Prof. Hary \_\_\_\_\_ from Australia yesterday. (Choose the suitable verb form.) has returned/ returned/ has been returning/ return back
- c. **Direct speech ⇌ Indirect speech** 2M L3 CO3  
 i) The teacher said to the student, "Do not waste time before your exam."  
 ii) I told my supervisor that I would be finishing the task the next day.
- d. **Active voice ⇌ Passive voice** 2M L3 CO4  
 i) Beena is humming a beautiful tune.  
 ii) The brinjal will be finely sliced by the chef.
- e. **Correct the given sentences.** 2M L3 CO5  
 i) Neither she nor her friends is interested in sports.  
 ii) Each of the girls have precipitated in the game.  
 iii) Salesman/Saleswoman (Find an inclusive term.)  
 iv) The little boy is **handicapped**. (Write a suitable euphemism.)
- f. i) Make **prudent** use of your time. (Choose a suitable synonym.) 2M L1 CO3  
 I. Judicious II. Mandatory III. Trivial IV. Imbecile  
 ii) Mr. Alex is a **pretentious** art critic. (Choose a suitable antonym.)  
 I. Flatulant II. Hollow III. Genuine IV. Overambitious  
 iii) The red rose tucked in your open hair \_\_\_\_\_ your saree.  
 (Compliments/Complements)  
 iv) Government by the people is known as \_\_\_\_\_. (Write the one-word substitute.)

## Part-B

4X12=48M

2. i) William Hazlitt feels that being in a school /hostel will teach his son about how to get along with others and prepare him for the ups and downs of life. Do you agree to the opinion? Analyse with textual references. 6x2M L4 CO1  
L2
- ii) Explain any four essentials to draft an effective paragraph.
3. i) "If you can meet with Triumph and Disaster  
And treat those two impostors just the same;"  
Who is the author of the poem? What is the meaning of the word 'imposter'? Analyse the significance behind labelling 'triumph' and 'disaster', 'two imposters.' 1+1+4 M L1, L4 CO2
- ii) Outline the differences between expository and descriptive writing style? 3+3 M  
Read the following paragraph and determine whether it is an expository or a descriptive paragraph. Give reasons to your answer. L2, L3  
The terrestrial planets are made of rock and metal and are closest to the sun. These include the midsize planets Mercury, Venus, Earth, and Mars. They rotate slowly and don't have many moons. Farther from the sun are the planets called gas giants, Jupiter, Saturn, Uranus, and Neptune. They are called gas giants because they are formed from gases such as hydrogen and helium.
4. i) Explain a few strategies to craft a persuasive paragraph. 6 M L2 CO3  
ii) What does the term "Dream" refer to in Martin Luther King's speech? Examine if this famous speech holds its relevance in today's time too. Justify your answer with relevant examples. 3 +3 M L4
5. i) "I took the one less travelled by, // And that has made all the difference." Who says this? Extend your views on the significance of these two lines. 6X2M L1, L2 CO4  
ii) Compose an argumentative essay on "Has the internet made our society better?" L6
6. i) Discuss any four characteristics of a successful blog. 4+ 8 M L2 CO5  
ii) Justify how writing on Instagram or Twitter is different from academic writing. Has writing on social media platforms impacted academic writing? Illustrate with examples. L5
7. i) As an effective communicator, why should you learn the uses of Inclusive language and Euphemism? Cite 3 examples for each. 6x2M L2 CO5  
ii) Develop a coherent paragraph by using the given hints: L6 CO1  
Giving pledge – Philanthropic cause – launched by Bill Gates– many are unprivileged – need support – the wealthiest people can share – make the world a better place – social responsibility





Hall Ticket No. -

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## Stanley College of Engineering & Technology for Women (A)

B.E (CSE/IT/CME/AI&DS) II Semester (Main) Examinations-September-2023

### Data Structures with C

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
  - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
  - Missing data, if any, may be suitably assumed.

#### Part-A

6X2=12M

- List any four applications of data structures. [2M CO1 BTL1]
  - Write a recursive algorithm for finding factorial of a number [2M CO2 BTL2]
  - The size of the stack is 7. The operations performed are push(35), push(40), push(45), pop(), push(50), pop(), push(55), pop(), pop(). Evaluate the final elements present in the stack. [2M CO7 BTL5]
  - What are the conditions for a balanced binary search tree? [2M CO9 BTL2]
  - Define Hash function. [2M CO9 BTL1]
  - Differentiate a tree and a graph. [2M CO12 BTL2]

#### Part-B

4X12=48M

- Illustrate asymptotic notations with examples. [6M CO1 BTL3]
  - Write the steps for Binary search algorithm and compare the time complexity with linear search. [6M CO5,12 BTL2]
- Evaluate the following infix expression using Stack:  
 $2 * (5 * (3 + 6)) / 5 - 2$  [6M CO7 BTL5]
  - Explain the implementation of queues using arrays. [6M CO8 BTL2]
- What are the applications of linked list data structure? [6M CO8 BTL2]
  - Compare circular linked list and doubly linked list. [6M CO7,8,12 BTL2]
- What are the variants of Binary trees? Explain. [6M CO9 BTL2]
  - Compare BFS and DFS with an example. [6M CO9 BTL2]
- Explain all the operations of m-way search trees. [8M CO10 BTL2]
  - What is linear probing? Explain. [4M CO9 BTL2]
- Sort the following numbers using quick sort:  
32,19,20,45,65,12,50,18 [6M CO4 BTL4]
  - Explain the rotations of AVL tree. [6M CO10 BTL2]

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Hall Ticket No. -

Code: R123929

**Stanley College of Engineering & Technology for Women (A)**

**B.E (IT) II Semester (Main) Examinations-September-2023**

**Applied Physics**

**Time: 03 Hours**

**Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.  
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iii. Missing data, if any, may be suitably assumed.

**Part-A**

**6X2=12M**

1. a. List any four applications of Laser [2M CO1 L1]  
b. What are polar and nonpolar dielectric materials? [2M CO2 L1]  
c. Explain Meissner effect. [2M CO3 L2]  
d. Classify intrinsic and extrinsic semiconductors [2M CO4 L4]  
e. Distinguish between bulk materials and films. [2M CO5 L2]  
f. Calculate the numerical aperture (NA) and acceptance angle of an optical fiber having 1.48 and 1.46 as refractive index of core and cladding respectively. [2M CO1 L3]

**Part-B**

**4X12=48M**

2. a) Explain the construction and working of Ruby laser. [7M CO1 L2]  
b) Explain double crucible method to fabricate optical fibers. [5M CO1 L2]
3. a) Discuss the effect of frequency on various polarization processes. [6M CO2 L4]  
b) Explain Weiss molecular field theory of ferromagnetism. [6M CO2 L2]
4. a) Develop an expression for wave function and energy associated with a particle in 1D box. [7M CO3 L3]  
b) Classify types of superconductors. [5M CO3 L4]
5. a) Explain the salient features of Kronig-Penney model with graph. [6M CO4 L4]  
b) Explain the formation of P-N junction diode and Discuss V-I characteristics of diode [6M CO4 L2]
6. a) Explain sol-gel method to prepare nano materials. [6M CO5 L2]  
b) Illustrate the electron beam evaporation method to prepare thin films [6M CO5 L4]
7. a) Develop an expression for electronic polarizability in di-electric materials. [8M CO2 L3]  
b) Define Hall Effect and write its applications. [4M CO4 L2]

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